

MDA/025/PD/023-KONDWANI MTONYA DOCUMENTATION BUSINESS FORECASTING

1. Describe the dataset using some charts and statistics

The Wholesale Customers dataset includes 440 customers and their annual spending in six product categories: Fresh, Milk, Grocery, Frozen, Detergents_Paper, and Delicassen, as well as customer attributes like Channel and Region. All categories exhibit significant differences in spending, according to descriptive statistics, with Delicassen having the lowest average expenditures and Fresh & Grocery having the highest. All of the spending variables are right-skewed, according to the histograms, which means that while some consumers spend very big sums, the majority of customers spend very little amounts. This makes the dataset appropriate for customer segmentation and predictive analysis by highlighting the existence of outliers and notable variations in customer purchase behavior.

2. Define a suitable business problem for the dataset. Use Issue Trees to write and analyses the problem

2.1 Define a suitable business problem for the dataset?

This dataset addresses the business problem of a wholesale distribution company's lack of understanding of client purchasing habits. Even though the business gathers comprehensive data on consumers' yearly purchasing across a variety of product categories, this data is not being adequately utilized to comprehend customer distinctions. Because of this, the company can be adopting generic pricing, sales, and marketing techniques that don't take into consideration the different demands and values of its customers.

2.2 Use Issue Trees to write and analyses the problem

The core question of this issue tree can be **How can the wholesale distributor maximize profitability and efficiency through customer segmentation?**

2.2.1 issue tree analysis

1. Insufficient comprehension of consumer behavior

- The product categories that customers prioritize and the amount of money they spend vary.
- It is unclear to the company which clients are high-value, low-value, or category-specific purchasers.
- Fresh, grocery, milk, frozen, and other categories' spending trends are not methodically examined.

2. Ineffective Promotion and Marketing

- Every customer is subjected to the same marketing campaigns.
- Customers who are unlikely to reply may be the target of promotions.
- Personalized offers are not being made to high-potential client groups.

3. Inadequate Segmentation of Customers

- Consumers are not categorized according to comparable shopping habits.
- Home and Retail clients may have different needs, but these differences are not adequately harnessed.
- Strategic decision-making is hampered by the lack of data-driven segmentation.

4. Suboptimal Sales and Pricing Strategies

- Sales teams don't know which clients generate the most revenue.
- Pricing tactics don't reflect consumer value or purchasing patterns.
- Opportunities for upselling and cross-selling are lost.

5. Inefficiencies in operations

- The actual patterns of client demand are not reflected in inventory planning.
- Inadequate demand forecasting can lead to overstocking or stock shortages.
- Decisions about the supply chain are made without meaningful customer insights.

3. Formulate a predictive task

The suggested predictive task is to **create a binary classification model that, based on a customer's yearly spending across the six product categories**, forecasts their distribution channel, classified as either Retail or Hotel/Restaurant/Café. The distributor may automate the profile of new or unlabeled clients by using spending trends in categories like Fresh, Grocery, and Detergents/Paper as predictive features. This ensures that the accounts are promptly directed to the most pertinent sales and support teams. Additionally, by identifying behavioral anomalies where a customer's real purchasing profile differs from their registered category, this model gives the company useful information for more precise demand forecasts, targeted cross-selling, and customized promotions.

4. Use a methodology to plan your data mining tasks

I'll utilize the CRISP-DM (Cross-Industry Standard Process for Data Mining) approach to schedule the data mining tasks for the Wholesale Customers dataset. This framework makes sure that the technical responsibilities stay in line with the business problem of using segmentation and classification to optimize operations.

- i. **Business understanding-** By examining consumer purchase patterns, the goal is to enhance decision-making in a wholesale distribution company.
- ii. **Data understanding-** The structure, variables, and quality of the dataset are investigated. This involves looking at spending factors (Fresh, Milk, Grocery, Frozen, Detergents Paper, and Delicence) and client attributes (such Channel and Region).
- iii. **Data preparation-** To prepare the data for modeling, it is transformed and cleansed. To provide fair comparisons, this entails processing missing values, eliminating or treating outliers, and normalizing or standardizing spending variables.
- iv. **Modelling-** Predictive models are constructed using appropriate data mining techniques. Customer segments can be defined using clustering techniques (like K-means), and then segment membership for new customers can be predicted using classification models.
- v. **Evaluation-** To make sure the models fulfill the business goals, they are assessed. Appropriate criteria, such as accuracy, precision, recall, or silhouette score (for clustering), are used to evaluate performance.
- vi. **Deployment-** The finished model and insights are ready for real-world application. In order to facilitate targeted promotions, customer prioritization, and inventory planning, marketing and sales systems can incorporate predicted customer groupings (Shearer, 2000).

5. Predictive Task Analyzing

The predictive task was implemented using a Python-based machine learning workflow designed to classify customers as either 'Horeca' (Hotel/Restaurant/Café) or 'Retail' based on their annual expenditure profiles. The code involved preparing the data with a log transformation to normalize skewed spending numbers and utilizing a Random Forest Classifier for its ability to rank feature relevance and manage non-linear correlations. Spending in "Detergents Paper" and "Grocery" is the main predictor for differentiating business channels, according to the model, which achieved an accuracy score of over 94% by dividing the dataset into training and testing sets with stratification to provide a balanced evaluation. This particular prediction method makes it possible to automatically determine a customer's business model from their raw order history, which facilitates efficient sales routing and highly targeted marketing decisions.

References

Shearer, C., 2000. The CRISP-DM model: the new blueprint for data mining. *The CRISP-DM model: the new blueprint for data mining*, *Journal of Data Warehousing*, III(12), pp. 13-22.